Shrvaani S

AI/ML Python Developer | GEN AI | Product/Project Management | Data Science | Data Analysis

Phone: +91 87606 01342 | sshrvaani@gmail.com

GitHub: https://github.com/Shrvaani

LinkedIn: https://www.linkedin.com/in/shrvaani

Professional Summary

Certified Data Scientist and AI/ML Developer with strong experience building intelligent, data-driven applications using Python, Django REST Framework, PostgreSQL, Generative AI, and AWS. Skilled in deploying machine-learning models, engineering scalable REST APIs, and delivering end-to-end solutions across EdTech, Telematics, and Retail. Proficient in designing Retrieval-Augmented Generation (RAG) systems, AI agents with LangChain and LangGraph, and multi-step reasoning workflows using LLMs.

Well-versed in backend engineering, object-oriented programming, database design, and secure API development. Hands-on experience building automation workflows in n8n using core nodes, triggers, and data transformation logic. Adept at working in Agile environments and collaborating with engineering, design, and product teams to ship scalable, user-centric, and business-aligned AI products.

Technical Skills

- **Languages:** Python, JavaScript (ES6+), Java, SQL.
- **AI/ML:** TensorFlow, PyTorch, Scikit-Learn, Hugging Face Transformers, OpenCV, NLTK, GPT-based APIs, Gemini models, o3/o4-mini reasoning models, Prompt Engineering (basic), RAG, LangChain, LangGraph, Agentic AI Frameworks, Image Generation (DALL·E, Diffusers)
- **Web Frameworks:** Flask, Django, Streamlit, n8n, Railway.
- **Data Analysis & Visualization:** Pandas, NumPy, Matplotlib, Seaborn, Plotly
- **Databases:** MySQL, SQLite3, Supabase.
- **Tools:** Git, GitHub, Jira, Trello, VS Code, Xcode, Cursor
- **Cloud & DevOps:** AWS, GitHub Actions, REST APIs
- **Product & Collaboration Tools:** Jira, Trello, Notion, Figma

Work Experience

Software Trainee
Montbleu Technologies Pvt Ltd, Coimbatore
Oct 2023 – Jun 2024

- Developed and maintained backend APIs and real-time data pipelines for **MotivAI.tech** — an AI-based telematics product improving road safety and reducing fleet accident rates.

- Integrated **OpenCV with Flask** for real-time object detection, lane tracking, and driver behavior analytics to enhance end-user safety insights.

- Deployed **TensorFlow models** for predictive analytics on driver performance, contributing to product features that improved risk prediction accuracy.

- Collaborated with product and QA teams to translate user requirements into feature specifications and refine the product roadmap.

- Managed database architecture (MySQL/SQLite), asynchronous task queues, and API security for scalable, production-ready deployment.

- Supported **CI/CD workflows** and product releases through GitHub Actions, ensuring version control and continuous improvement cycles.

Product Development & Program Management Trainee
Iamneo (Formerly Examly), Coimbatore
Aug 2024 – Feb 2025

- Designed and implemented intelligent feedback systems using **OpenAI APIs**, improving learner engagement and personalization metrics.

- Defined product requirements in collaboration with stakeholders and engineering teams, aligning features with user insights and business goals.

- Designed backend logic in **Django** for dynamic assessments and data dashboards to support real-time progress tracking.

- Coordinated **Agile sprint planning**, QA testing, and UX enhancements using Jira, ensuring smooth product delivery and iteration cycles.

- Partnered with design and content teams to enhance the user journey and streamline onboarding flows across the learning platform.

- Monitored product KPIs (engagement rate, test completion time) to inform feature prioritization and iterative improvements.

Key Projects

AI Video Analysis System (YOLOv8 | DeepFace | TorchReID | Streamlit | OpenCV | Python)

Deployment Link: https://ai-video-analysis-system.streamlit.app/

Github Link: https://github.com/Shrvaani/AI-Video-Analysis-System

- Developed a **real-time AI video analytics platform** using **YOLOv8**, **DeepFace**, and **TorchReID** to detect, track, and re-identify people in CCTV footage. Designed to enhance security and retail insights by identifying unique individuals, logging entry–exit times, and tracking repeated appearances across videos.
- Built an **interactive Streamlit dashboard** that visualizes detections with bounding boxes, tracks analytics, and logs entry/exit data automatically: integrated **MD5 video hashing**, **GPU acceleration**, and optimized frame processing for faster, more reliable inference.
- Delivered a **scalable, end-to-end AI solution** that combines computer vision, data analytics, and user experience—demonstrating strong technical execution, product design thinking, and deployment-ready implementation.

Accelerometer Activity Classifier (Python | Streamlit | Random Forest | Data Visualization)

Deployment Link: https://accelerometer-activity-classifier.streamlit.app/

Github Link:
https://github.com/Shrvaani/Accelerometer-Activity-Classifier/tree/Accelerometer-Activity-Classifier

- Developed a **machine learning web application** using **Python** and **Streamlit** to classify human physical activities from raw accelerometer data. Designed preprocessing pipelines for handling missing values, feature engineering, and window-based segmentation of tri-axial sensor readings (X, Y, Z).
- Trained and optimized a **Random Forest model** to classify eight distinct activities, achieving high accuracy on time-series test data. Integrated **interactive visualization**, CSV/Excel data upload, and batch prediction features, allowing users to explore insights dynamically.
- Deployed a user-friendly interface for real-time inference and data exploration, focusing on usability for **fitness tracking, behavioral analytics, and research applications**.

AI Assistant (RAG Chatbot using Django + Railway Deployment)

Deployment link: https://rag-chat-assistant.up.railway.app/

Github link: RAG_Chatbot

- Built a production-grade RAG backend with **Django** and **OpenAI** (Chat/Completion APIs). Implemented RESTful endpoints for chat, document upload, and retrieval with clear request/response contracts. Designed stateless, scalable handlers to deliver low-latency, real-time conversational responses.
- Engineered vector search using **sentence-transformers** for embeddings and a Chroma/FAISS-compatible store. Implemented top-k semantic retrieval, relevance scoring, and inline citation injection into responses. Tuned embedding + retrieval pipeline for precision and low hallucination risk in document-grounded answers.
- Built a secure document ingestion pipeline (PDF/DOCX/TXT → parsing → chunking → embedding). Added sanitization, file-size limits, metadata tracking, and persisted chat & document metadata in **PostgreSQL**. Enabled session-aware chat history and reliable context rehydration for multi-turn conversations.
- Deployed as a scalable service on **Railway** using Docker-friendly CI/CD and environment-managed builds. Integrated an admin dashboard, structured logging, and basic observability for performance and incident

triage. Architected for horizontal scaling and production readiness with clear operational runbooks.

AI-Driven ML & Analytics Projects

Instagram Reach Analytics (Python, Plotly, Streamlit)

Deployment Link: https://instagram-reach.streamlit.app/

Github Link:
https://github.com/Shrvaani/DataPyHub/tree/Python_Analyzing-Instagram-Reach

- Built an automated analytics pipeline using Python (Pandas, TextBlob) to clean Instagram post data, generate sentiment scores, engineer engagement-driving features, and uncover performance trends.
- Developed interactive visual dashboards in Streamlit, resolving Arrow/dtype rendering issues and enabling real-time insights into impressions, engagement rate, caption strategy, and hashtag effectiveness.
- Delivered a modular, production-ready analysis framework used for content performance optimization and data-driven decision-making.

Weather Prediction System (ML Regression, Data Pipelines)

Deployment Link: https://weather-prediction-py.streamlit.app/

Github Link:
https://github.com/Shrvaani/DataPyHub/tree/Python_Based_Weather_Prediction

- Built a supervised regression model to predict temperature and humidity using Scikit-learn, implementing feature scaling, transformation, and rigorous train-test validation.
- Structured a clean ML pipeline covering ingestion, preprocessing, model training, evaluation, and artifact storage using `joblib` for reproducible inference.
- Developed a deployable inference script enabling real-time weather prediction from new input values, supported by model-performance charts and error diagnostics.

IPL Match Prediction Model (ML Classification, Feature Engineering)

Deployment Link: https://ipi-prediction-python.streamlit.app/

Github Link:
https://github.com/Shrvaani/DataPyHub/tree/Python_Based_Analysis_of_IPL

- Engineered a full data-processing workflow for 2008–2024 IPL datasets, performing merging, normalization, cleaning, and feature extraction (toss impact, venue patterns, win metrics) from match + ball-by-ball data.
- Trained and deployed a machine-learning model to predict match outcomes using encoded categorical features, and integrated model/encoder artifacts for cloud-based inference.
- Developed an interactive Streamlit dashboard showcasing team trends, player stats, toss analytics, and live ML predictions, with fully versioned model files and reproducible deployment setup.

Hugging Face Projects:

AI Assistant – LLM-Based Conversational Agent (Python, HuggingFace Inference API, Streamlit)

Deployment Link: https://ai-assistant-huggingface.streamlit.app/

Github Link:
https://github.com/Shrvaani/Gen-AI-Using-Python/tree/AI_Assistant

- Built a Python service layer that interfaces with HuggingFace LLM endpoints using token-efficient request batching, structured prompts, and asynchronous API execution.
- Built a Streamlit front end that renders model outputs in real time, using modular prompt handlers and state management for consistent conversational context.

AI-Powered Image Gallery – Vision-AI Metadata Pipeline (Python, HuggingFace Vision Models, Streamlit)

Deployment Link: https://ai-powered-image-gallery.streamlit.app/

Github Link:
https://github.com/Shrvaani/Gen-AI-Using-Python/tree/AI_Powered_Image_Gallery

- Implemented an automated vision pipeline using CLIP/ViT models to extract embeddings, generate captions, and derive semantic tags for image indexing.
- Developed a model-driven search interface in Streamlit that performs embedding-based retrieval, dynamic filtering, and metadata rendering.

Tic Tac Toe – AI-Powered Multiplayer Game (FastAPI, React, Firebase, Railway, Vercel)

Deployment Link: https://tic-tac-toe-ten-topaz.vercel.app/

Github Link: https://github.com/Shrvaani/Tic_Tac_Toe

- Built a distributed full-stack architecture using FastAPI for backend APIs, React for the client interface, and Firebase Realtime Database for persistent multiplayer state, session sync, and leaderboard updates—supporting real-time gameplay across concurrent users.

- Implemented an AI-powered Minimax engine for optimal single-player moves and deployed a fully decoupled system using Railway (backend) and Vercel (frontend) with CI/CD integration for scalable, production-grade performance.

Education

Bachelor of Science in Information Technology
Nirmala College for Women, Coimbatore (2020-2023)

- **Grade**: A+ | Affiliated to Bharathiar University
- **Awarded:** Best Student of the Year (2020–2023)
- **Leadership**: **Department Secretary;** represented the college in technical and non-technical events

---

Certifications

- Advanced Diploma in Data Science, AI & Machine Learning (2023)

Languages

- English: Full Professional Proficiency

- Tamil: Native/Bilingual Proficiency

Career Vision

- To become a proficient AI/ML Software Developer, one must create scalable, intelligent systems.

- To actively contribute to open-source and tech communities.

- To explore advancements in cloud computing, big data, and blockchain for real-world impact.

- To drive business outcomes with AI/ML, predictive analytics, and automation at scale.
